



Instructor: Date:

Learner: Regional Branch:

Program: Center:

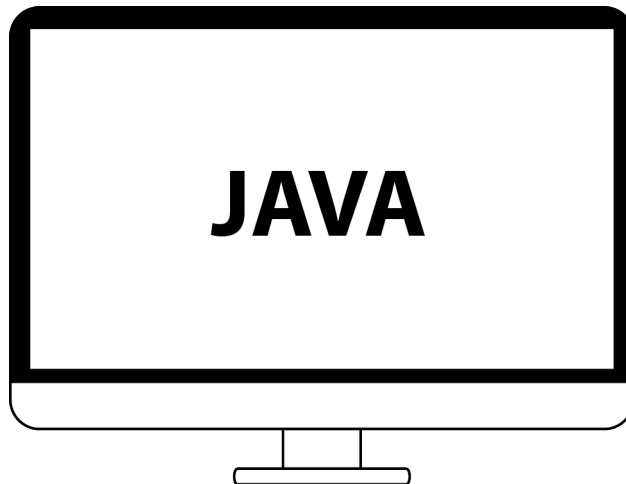
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Let's Start!

What operators does Java work with?



Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- When do you think you have to use arithmetic operations in Java?
- What Java functions do you know or have you worked with?

Can Do Statements:

- I can list the assignment operators used in Java.
- I can recall the syntax to use for different assignment operators in Java.
- I can talk about shortcut operators and their use in Java.





Let's Explore!

Catherine and Louis go over Java-Assignment Operators.

Assignment Operators



Instructions: Read the text below.

- Louis:** Hey Catherine! I'm supposed to hand in a summary about Java assignment operators next week but I barely remember what they are. I used to work with Java very often, but now, I feel I don't understand most of its functions.
- Catherine:** Java isn't supposed to be difficult. These assignment operators include mathematical operators that are used to give values to variables. If you would like, I can explain to you how these operators work and their syntax.
- Louis:** That'll be great!
- Catherine:** Alright. The first two operators are Addition and Subtraction. As their names suggest, we use these operators to add or subtract value to a variable. I used to work a lot with these operations so I can give you a clear example. Imagine you have three variables with different values: `int a = 10`, `int b = 5`, and `int c = 1`. Now we perform the following mathematical operation: `a = b + c`. Here, the value of `a` is supposed to change from 10 to 6. We would follow the same process when we subtract.
- Louis:** I think I remember this. Did you use to work with Multiplication and Division operations too?
- Catherine:** For sure! These are supposed to be very common in Java. Let me give you a similar example. Take the following variables: `int q = 15` and `int t = 0`. Now let's perform the operation `t = q/3`. Here, the value of `t` will change from 0 to 5. However, you have to keep in mind that in Division, we can get a remainder when the result is not an even number. For instance, if I perform the following operation `int n = 25/3`, I will have a remainder equal to 1.
- Louis:** I think I didn't use to work that much with divisions. I don't remember ever having a remainder. I know there are also shortcuts in Java, would you help me remember what they are?
- Catherine:** Yes. These are operators which you can use to add or subtract 1 from a number. For example, we would use Increment and Decrement when we want to quickly increment or decrement a number by 1. If you want to increase or decrease the value of a variable by a value different from `1`, use Plus-equals or Minus-equals.
- Louis:** I would definitely put this into practice. Thank you Catherine!



Let's Practice!

Different Assignment Operators

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. I perform the operation `'g = f + 3'` with the variables `'int g = 10'` and `'int f = 3'`.
2. I perform the operation `'t = 21/e'` with the variables `'int t = 3'` and `'int e = 7'`.
3. Usually, when I work with this operator, I get a remainder.
4. I take away a specific value from a variable. This process is called...
5. If I have the variable `'r'` with a value equal to 6 and I decide to add that value 5 times, what operation am I performing?
6. I perform the operation `'k = 4 - r'` with the variables `'int k = 5'` and `'int r = 1'`.

These sentences refer to Addition and Subtraction operators	These sentences refer to Multiplication and Division operators



Let's Practice!

Mr. Thomson's Classes

Instructions: Read the paragraph. Write the missing word(s) in the blank. Use: used to believe, used to insist, used to make, used to fail.

Mr. Thomson _____ on the importance of the differences and functions of Java operators. However, I _____ that shortcut operators had the same functions as assignment operators, so I _____ all the exams on this topic; nevertheless, he _____ different workshops in class to help me understand that these two operators are not the same. Fortunately, I was able to understand this topic as I now have to use these operators very often for the C# course I'm taking this semester.



Extension Activity

Operators in Java

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about assignment operators in Java.
- Make a poster in which you explain some characteristics and important details about the operators that you chose. Give some examples.
- Share your poster with the rest of the class.

Self-Assessment

Yes

No

- I look for information about other assignment operators in Java.
- I make a poster, explaining important characteristics them
- I share the poster with the rest of the class.

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Target Words

- hand in** (phrasal verb): to give something to someone
e.g. I'm supposed to hand in a summary about Java assignment operators next week but I barely remember what they are.
- syntax** (noun): it is a set of rules that indicate how the elements in a system should interact
e.g. If you would like, I can explain to you how these operators work and their syntax.
- perform a mathematical operation** (expression): to carry out the action of adding, subtracting, multiplying, etc.
e.g. We perform the following mathematical operation: $a = b + c$.
- remainder** (noun): a number or value that is left over
e.g. You have to keep in mind that in Division, we can get a remainder when the result is not an even number.
- shortcut** (noun): the fastest way of form to accomplish something
e.g. I know there are also shortcuts in Java, would you help me remember what they are?
- put something into practice** (expression): to practice or try something that you recently learned
e.g. I would definitely put this into practice.



Learning Points

Used to

We use **Used to** for past actions that happen frequently in the past.

Let's take a look at the following structure:

Subject+ **used to**+ verb in base form + complement

Example:

*I **used to** teach assignment operators in class.*

In the negative form, we need to add the auxiliary *didn't*. So, the verb **used** takes the base form.

Example:

*I **didn't use to** pass the exams of shortcut operators at university.*

Similarly, when we want to use the interrogative form of the expression used to, we also have to use the auxiliary did.

Example:

***Did** you **use to** work with java when you were in college?*

Note: Be careful not to mistake **used to** with **be used to** or **get used to**; these other forms represent a habit in the present or a habit that someone is acquiring.

Examples

- *I used to work with Java very often, but now, I feel I don't understand most of its functions.*
- *Did you use to work with Multiplication and Division operations too?*
- *I think I didn't use to work that much with divisions.*



Learning Points

Assignment Operators and Shortcuts

Assignment operator is the operator used to assign a new value to a variable.

Assignment operators is a kind of operator used in Java to manipulate variables. We can find the following ones:

a = b	Right operand is equal to left operand
A += b	A is equivalent to a + b.
A -= b	A is equivalent to a -b.
A *= b	A is equivalent to x times b.
A /= b	A is equivalent to b/a.
A %= b	A is equivalent to b%a.